

The Blockchain Sustainability Debate

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Speaker introduction



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Moritz is a Customer Engineer at Google Cloud in Berlin, where he focuses on supporting the German capital markets. In his role at Google, he collaborates with local teams to identify business opportunities and technical requirements, works on Google Cloud Platform integrations, and leads technical projects.

Concurrently, Moritz is a part-time PhD student in the Department of Informatics at King's College London, researching consensus protocols under the supervision of Prof. Peter McBurney. He holds degrees in information systems and computer science from TU Berlin and Birkbeck, University of London.

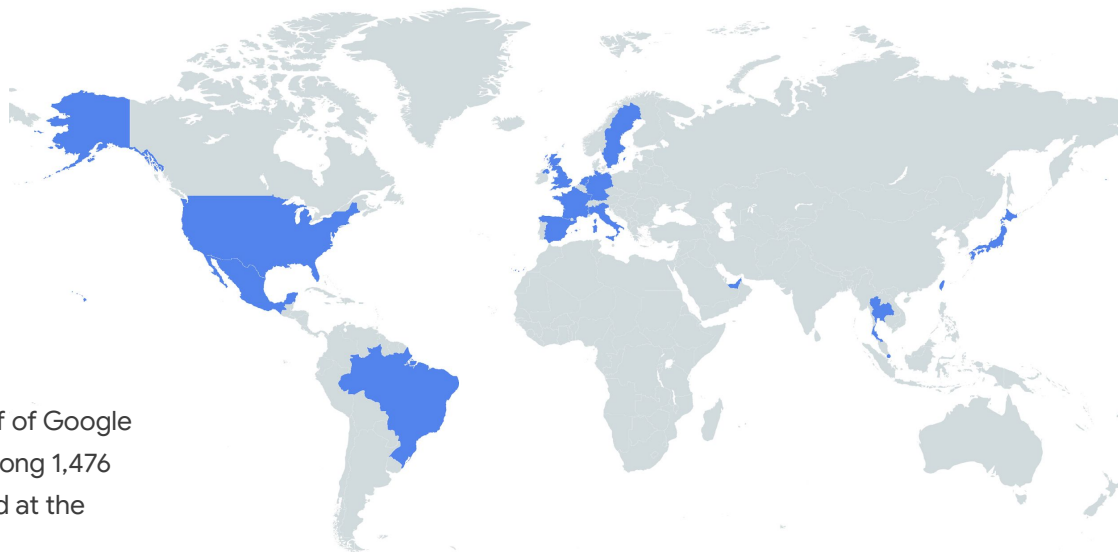
Sustainability: Still a Priority?	01
Blockchain's "Bad Rap"	02
Energy Consumption Drivers	03
Proof-of-Work and Alternatives	04
Addressing Environmental Concerns	05

Google Cloud Sustainability Survey

This study was conducted by The Harris Poll on behalf of Google Cloud from January 12, 2023, to January 23, 2023, among 1,476 executives across 16 markets. All respondents worked at the C-Suite or VP level.

All respondents worked in one of the following industries:

**Financial services; Retail/CPG;
Healthcare or life sciences;
Manufacturing or heavy industry;
Technology, Telecommunications;
Media, Entertainment, or Gaming;
Supply chain and logistics.**



All respondents worked in one of the following geographies:

US, Mexico, Canada, Brazil, UK, Australia, France, Japan, Italy, Taiwan, Germany, Thailand, Netherlands, Spain, Singapore, Sweden, UAE

Economic headwinds are stalling progress

45%

“The current **economic climate** is regressing sustainability efforts in my organization.”

67%

“The current economic environment is pushing my organization to **cut corners** in our sustainability initiatives.”

42%

European executives share that the **energy crisis** also affected their sustainability initiatives negatively.

But participation still remains high

Nearly all organizations (**96%**) have at least one program in place to advance sustainability initiatives, with the average company having four, which is consistent with last year.

Programs to advance sustainability initiatives

47%

Implements sustainable office policies

45%

Plans to reduce energy / migrate to renewable energy

44%

Supports green vendors

43%

Sustainable mindset in product / service design

42%

Makes an effort to offset carbon footprint

42%

Runs a recycling program

37%

Encourages sustainable transportation

33%

Donates services to green initiatives

27%

Donates funds to green initiatives

26%

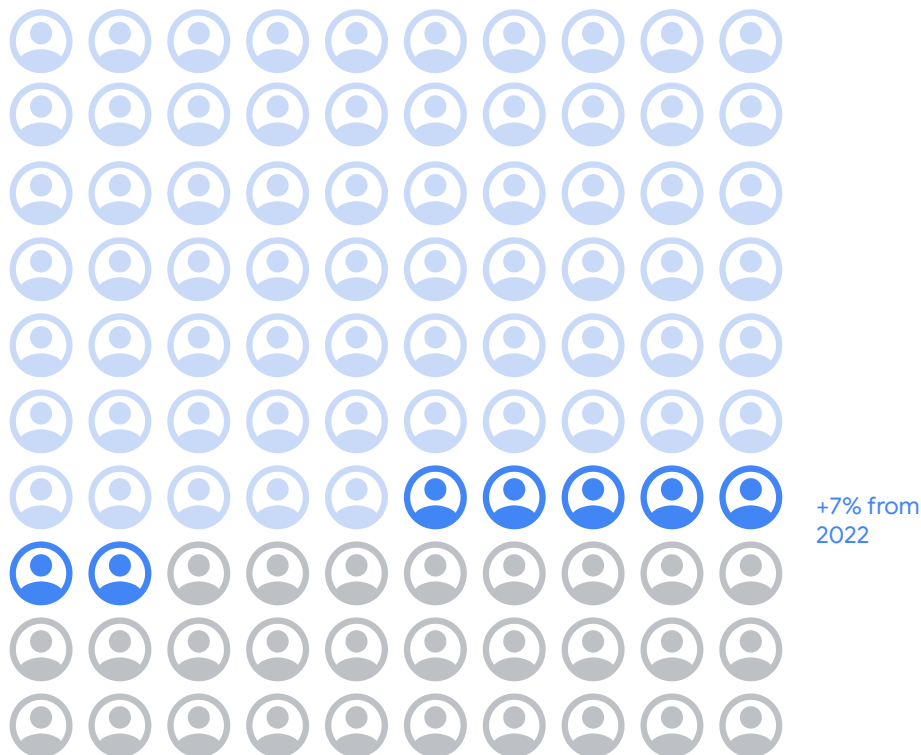
Volunteer program including green initiatives

4%

Do not have any sustainability programs

CEOs struggle to turn ambition into action

72% of executives agree they want to advance sustainability efforts, but no one knows how to actually do it.



Nine in 10 executives want their organization to become net zero emissions or have already met their goals



Q216: Where is your organization standing in terms of becoming a net zero emissions organization? (n=1,476)

01

Blockchain's “bad rap”



We determined electricity consumption to be between 60 and 125 TWh per year for Bitcoin. This is in the range of [...] Austria (75 GWh) and Norway (125 GWh)."

Sedlmeir, J., Buhl, H.U., Fridgen, G. et al.

[Sedlmeir, J. et al. 2020. The Energy Consumption of Blockchain Technology: Beyond Myth. Business & Information Systems Engineering. Springer.](#)

Blockchain Sustainability Perception

“Why Blockchain, NFTs, And Web3 Have A Sustainability Problem”

Bitcoin consumes 150 terawatt-hours of electricity every year: equal to the entire country of Argentina. This emits approximately 65 megatons of CO₂ annually.

Bernard Marr (Forbes)

<https://www.forbes.com/sites/bernardmarr/2023/01/13/why-blockchain-nfts-and-web3-have-a-sustainability-problem/>

“Where Cryptocurrency, Water and Conflict Collide”

The increased energy demand for Bitcoin mining tipped the grid over and power shortages led to localized blackouts in parts of the country, exacerbating existing tensions.

Kayly Ober (USIP)

<https://www.usip.org/publications/2023/03/where-cryptocurrency-water-and-conflict-collide>

“Bitcoin is terrible for the environment – can it ever go green?”

There is a recent push by some environmentalists to reduce the environmental impact of Bitcoin by changing the way it is produced.

Erum Salam (The Guardian)

<https://www.theguardian.com/technology/2023/apr/26/bitcoin-mining-climate-crisis-environmental-impact>

02

Energy Consumption Drivers

Payment System Components

	Traditional payments			DLT-based digital currencies		Non-DLT digital currencies
	Cash	Bank transfers	Credit cards	Permissioned	Permissionless	Centralized
Core processing	Creation, distribution, use, and disposal of banknotes and coins	Banks' data centers, RTGS	Card issuers' and banks' data centers, RTGS	Accredited nodes	Any node that joins the network to validate transactions	Central servers
User payment means		Website, cheques, bank branches	Physical cards and point-of-sale terminals	Digital wallets		Digital wallets

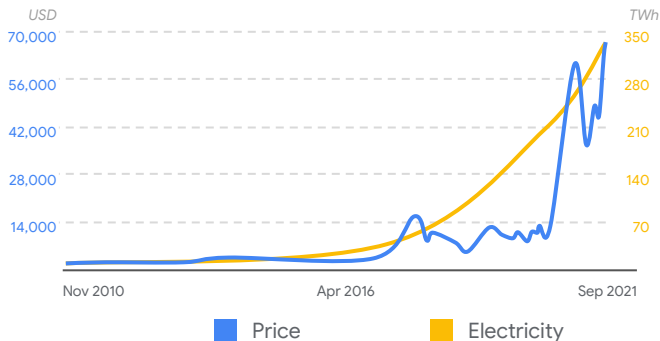
Agur, I. et al. 2023. Lessons from crypto assets for the design of energy efficient digital currencies. Ecological Economics. Elsevier.

Why does Proof-of-Work consume so much energy?

- In Proof-of-Work, validators (“miners”) receive rewards and fees in cryptocurrency
- Proof-of-Work uses a cryptographic puzzle to select the miner for the next block
- Difficulty of the puzzle is adjusted so that the block time is constant
- Whoever provides the solution first earns the reward
- The more computational power is invested into solving the puzzle, the higher the likelihood to be selected

Simple Economics of Proof-of-Work Mining

A miner will break even once the rewards and fees cover the costs for mining cryptocurrency. Their expenses are likely incurred in fiat currency, therefore the cryptocurrency price significant impact on mining expenses. A large proportion of mining expenses are electricity but there are others like hardware, personnel, facilities, etc.



Maiti, M. 2022. Dynamics of bitcoin prices and energy consumption. Chaos, Solitons & Fractals: X. Elsevier.

Rewards + Fees

Distributed according to Blockchain Protocol

× Cryptocurrency Price

Price in the currency in which mining equipment and electricity are purchased

≥ Mining Expenses

Costs a miner incurs for operating mining hardware

03

Proof-of-Work and Alternatives

Blockchain System Characteristics

Networking Complexity

Network permissioning has a great influence on energy consumption. Permissionless networks have the potential to grow large once they gain popularity.

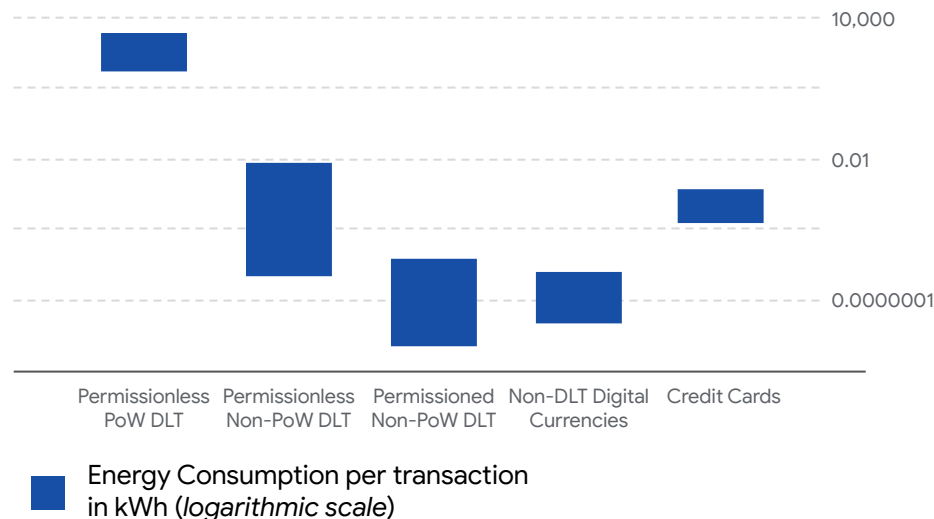
Ensuring decentralized consensus in permissionless networks is hard.

	Permissioned	Permissionless
Private	Central admissions process, centrally managed validators • Canton Network • Onyx	
Public	Anyone can join, centrally managed validators • Ripple • Hedera Hashgraph	Anyone can join, anyone can be a validator • Bitcoin • Ethereum

Not all Blockchains are Created Equal

Proof-of-Work is an outlier

The energy consumption of Proof-of-Work with difficulty adjustment is already very high under current market conditions. Energy consumption is sensitive to price changes.



Agur, I. et al. 2023. Lessons from crypto assets for the design of energy efficient digital currencies. Ecological Economics. Elsevier.

Ensuring Sustainability: External and Internal Factors

Regulators

Prohibitive Regulation

Impose bans on cryptocurrency mining (e.g. NY state Assembly Bill A7389C)

Fiscal Interventions

Asking miners to pay a premium for electricity (e.g. Digital Asset Mining Energy (DAME) tax equal to 30%)

Sustainability Criteria

Imposing minimum ESG criteria for institutions

Consumer Education

Making end users aware of the energy footprint of cryptocurrencies (e.g. energy efficiency labelling)

Organisations

Consider Electricity Consumption

Consider electricity consumption of protocol as a key criterion for chain selection

Run on the cleanest cloud in the industry

Google Cloud matched 100% of renewable energy globally in 2022

Enable developers to build sustainably

Google Cloud provides customers with carbon emission information

04

Achieving Sustainability with Google Cloud Platform Offerings for Web3

Google's efficient data centers help you meet your sustainability goals



Green IT: The cleanest cloud in the industry

Zero annual net operational carbon emissions on Google Cloud (since 2017)

Operate on 24/7 carbon-free energy, everywhere, by 2030

Net-zero emissions across our value chain + water consumption targets by 2030



Green with IT: Scale up sustainability solutions

Reduce your digital CO2 footprint - [Carbon Footprint](#) ; [Active Assist](#)

Understand and act on climate risk exposure - [Google Earth Engine solutions](#)

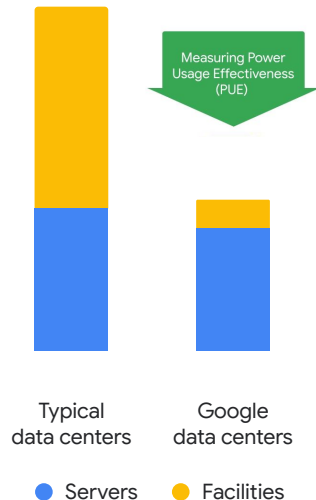
Learn about sustainable IT choices - [Sustainable IT. Decoded](#) masterclass

2X

Google data centers are twice as **energy efficient** compared to a typical enterprise data center

7X

as much computing power with the same amount of electrical power, compared to 5 years ago



Blockchain Node Engine

Blockchain infrastructure (most commonly nodes) deployed for external customers or internal Google Cloud projects require either significant manual effort to deploy, or development of custom automation solutions.

We will develop and make available a solution for automated deployment and maintenance of nodes. Development deliverables will include:

- Deployment automation for L1s and major L2s
- API, CLI, and UI for launching nodes-as-a-service
- Self-healing
- Fast-provisioning



Google Cloud

Blockchain Analytics

BigQuery blockchain public datasets already provide Bitcoin, Ethereum, and other blockchain datasets for consumption. These datasets have been some of the most popular of all Google Cloud hosted public datasets since their launch. However, various contributors—both internal and external—manage and maintain this work without an SLA/SLO or official RTO/RPO plans.

We will bring this offering in line with Google Cloud processes and policies, and enable a future project to make it a fully supported product. Milestones include:

- Establish SLAs/SLOs
- Historical raw data via BigQuery
- Real-time raw data via Pub/Sub
- Enrich datasets with smart contract and NFT metadata

The Google Cloud logo, featuring the word "Google" in its multi-colored font followed by the word "Cloud" in blue.

```
> web3.eth
.getBlock(1212121, true)
{ difficulty:
'20632001790827',
extraData:
'0xd7830103058447657468
87676f312e352e31856c696
e7578', gasLimit:
4712388, gasUsed:
409654, hash:
'0xe58280354d08c9fef9
```

Google Cloud's compliance offerings specific for Financial Services

Our products regularly undergo **independent third-party audits** with over 2 million control instances audited annually. Google maintains certifications, attestations of compliance, or audit reports against standards and regulations around the world.

Global



ISO/IEC 27001
ISO/IEC 27017
ISO/IEC 27018
ISO/IEC 27110
ISO/IEC 27701
ISO/IEC 9001
ISO/IEC 22301
ISO/IEC 50001
SOC 1 | 2 | 3
PCI DSS
PCI 3DS
CSA STAR

Americas

USA

Sarbanes- Oxley
Regulation SCI
SEC Rule 17a-4(f)
CFTC Rule 1.31(c)-(d)
FINRA Rule 4511(c)
FFIEC
FDIC
OCC
Federal Reserve Guidance
GLBA
CCPA
COPPA

Canada

PIPEDA
OSFI Guideline B-10

Brazil

LGPD
CMN4,983
BCB No.85

Argentina

PDPL
BCRA

Mexico

CNBV LIC | CUB
CNBV LMV | CUCB
CNBV CUF | Fintech Law
CNSF LISF | CUSF

Peru

Superintendencia de Banca

Colombia

Financial Superintendence rules

Chile

CMF

Europe, Middle East & Africa

Europe

GDPR
EU Standard Contractual Clauses
EU CoC
SWIPO
EBA
EIOPA
ESMA

Germany

BaFin / BAIT
MaRisk
KRITIS

Austria

BWG
VAG

Switzerland

FINMA
ISAE 3000 Type 2

Netherlands

DNB

Spain

Banco de España Circular 3/2022
Banco de España Circular 2/2016

South Africa

POPI

UK

FCA FG16/5
FCA SYSC 8
PRA SS1/21 & SS2/21

Ireland

CBI

France

ACPR

Italy

Banca D'Italia Circular 285

Portugal

Banco de Portugal

Luxembourg

CSSF

Poland

KNF

Sweden

SFSA

Denmark

FSA Executive Order 877

Turkey

BRSA

Asia Pacific

Australia

APP
APRA CPS 231
APRA CPS 234

New Zealand

Reserve Bank BS11
Privacy Act

India

RBI
SEBI
IRDAI

Malaysia

PDPA
BNM RMIT & Outsourcing

Indonesia

PDP Law
SEOJK 21
POJK 38
SNI 27001

Taiwan

PDPA
FSC Banking Outsourcing
FSC Insurance Outsourcing

Philippines

PDPA
BSP Manual of Operation
BSP ITRM

Japan

FISC
My Number Act
APPI

Korea

K-ISMS
FSC
PIPA

Singapore

MTCS Tier 3
OSPAR
MAS Guidelines
ABS Guide
PDPA

Hong Kong

HKIA Outsourcing GL14
HKMA Outsourcing SA-2
PDPO

Thailand

BOT FPG 19/2599 & 21/2562
OIC
PDPA

Vietnam

State Bank of Vietnam

Additional compliance offerings and further information is available at cloud.google.com/security/compliance

Take-Home Message

Sustainability is a growing concern throughout all industries with FSI being less accomplished.

Some Blockchain protocols consume a lot of electricity. Others don't.

Proof-of-Work is anti-efficient in terms of energy consumption.

There is regulatory interest on highly consumptive cryptocurrencies.

Google Cloud Platform can help enterprises achieve sustainability goals in Web3-exposed organizations.





Thank you.